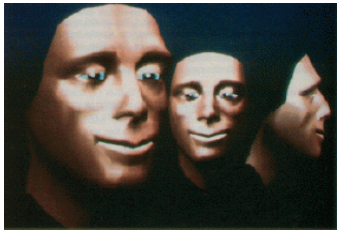
Data Driven Computer Animation

Facial animation

Taku Komura

Facial Animation

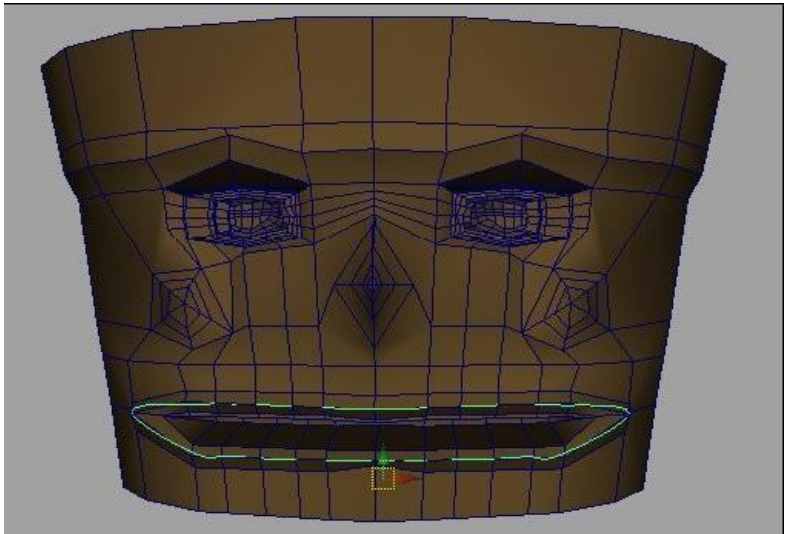
- Previously we have covered topics about full body motion, crowd motion etc.
- Facial animation is also one of the most important topics in character animation
- A lot of technique have been introduced in to help animators produce realistic facial animations
- A lot of data-driven techniques have been introduced

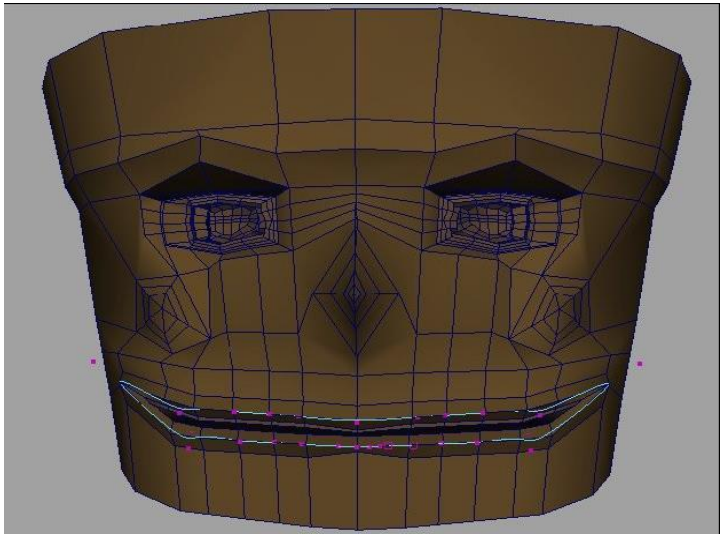






Facial Animation – why difficult?

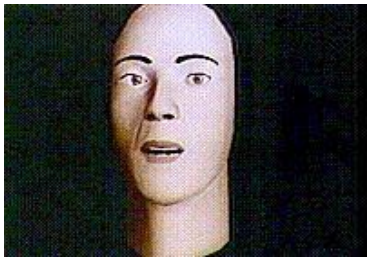
- The number of points that compose the face is large – the dimensionality is high
- Can be very tedious and time consuming to manually produce each expression without an underlying structure

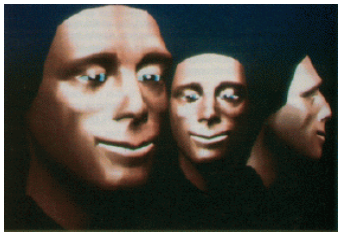




Overview

- Parke's parametric face
- Muscle-based models
- Deformation Transfer











Parke's Parameteric Model

Fred Parke created the first 3D parametric model of a human face.

The model is discussed in his 1974 dissertation.

A parametric model for human faces

Frederic Parke, 1974 Ph.D. dissertation, Utah

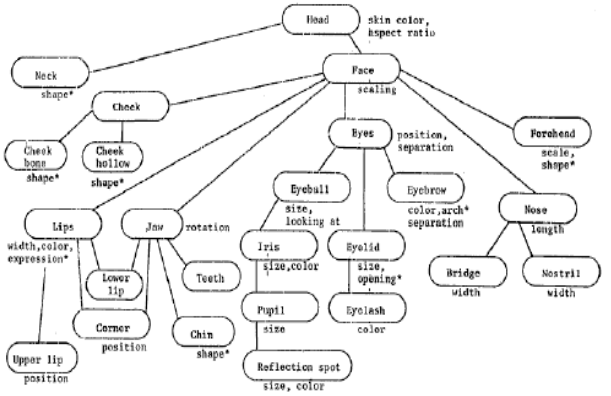
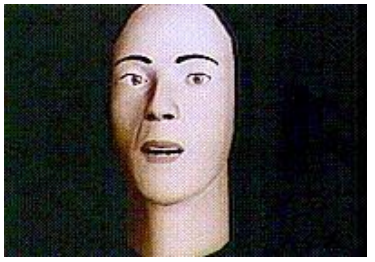


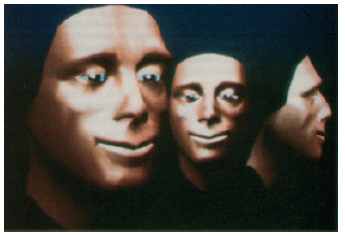
Figure 3.1 - The structure of the parametric model. The parameters affecting the various nodes are shown. An * indicates the use of interpolation to implement the parameter.



Overview

- Parke's parametric face
- **Muscle-based models**
- Expression cloning
- Deformation Transfer





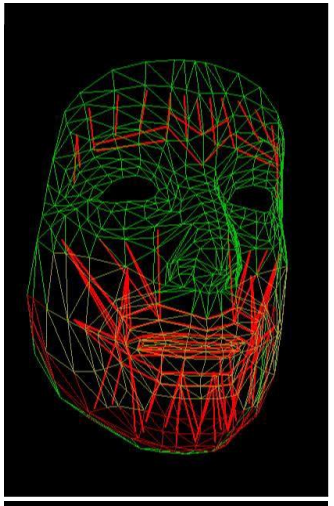




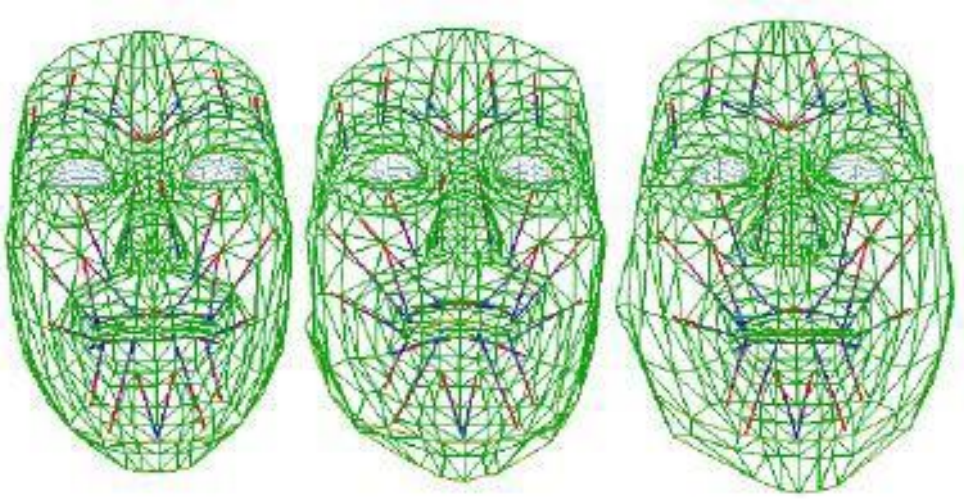


Muscle Based Animation

- Uses a mass-and-spring model to simulate facial muscles.
- Muscles are of two types: linear muscles that pull and elliptic muscles that squeeze.

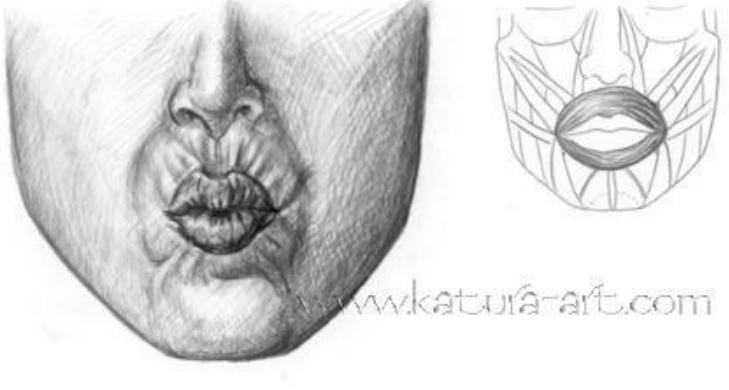


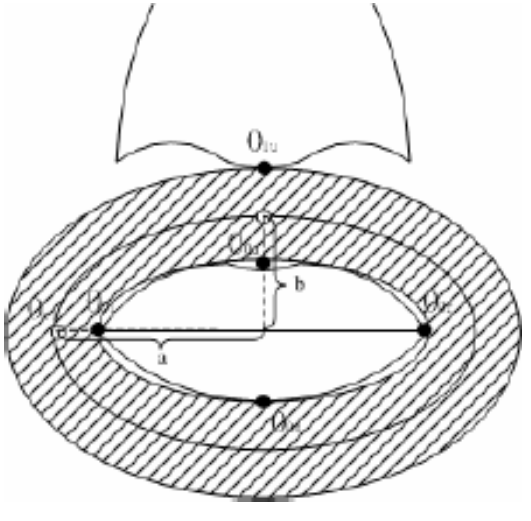
Polygonal surface



Elliptic Muscles

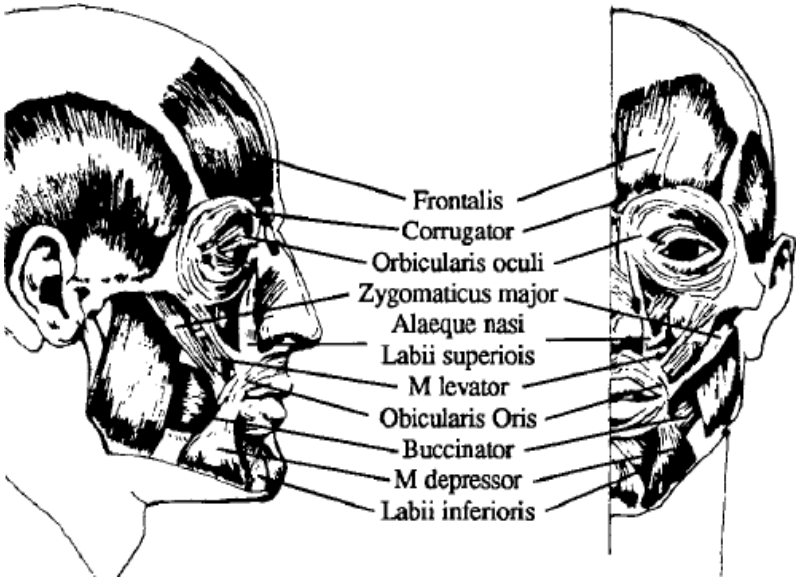
- Around the mouth (**Orbicularis oris)**
- The fibers encircle the mouth
- When contracted, it squeezes the mouth
- Functions: pressing together, tightening and thinning, a rolling inwards between the teeth, and a thrusting outwards





How to control the muscles?

- There are more than sixty muscles on the face
- There can be thousands of combination of activating the muscles
- Which muscles should be activated in which mood?



How to control the muscles (2)?

- Facial Action Coding System (FACS) :
  Ekman ‘77

- A widely used notation for the coding of facial articulation.
- 66 muscle actions and the resulting effects
- Describes the action units (muscles) involved in the six basic expressions

- Basic facial expressions that are considered to be generic to the human face:
  → *Happiness, Anger, Fear, Surprise, Disgust, Sadness*

Synthesized Facial Expressions
Waters SIGGRAPH '87



Neutral face

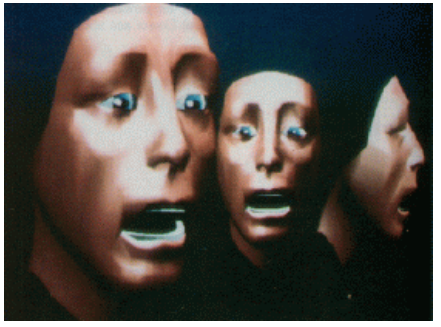
Anger





Happiness

Surprise





Fear

Disgust



Realistic models

- If we want to simulate the expressions of real humans, we need to scan the surface of faces

We need the

- **Geometry Data**
- **Texture**
  of the face





Cyberware Color Digitizer

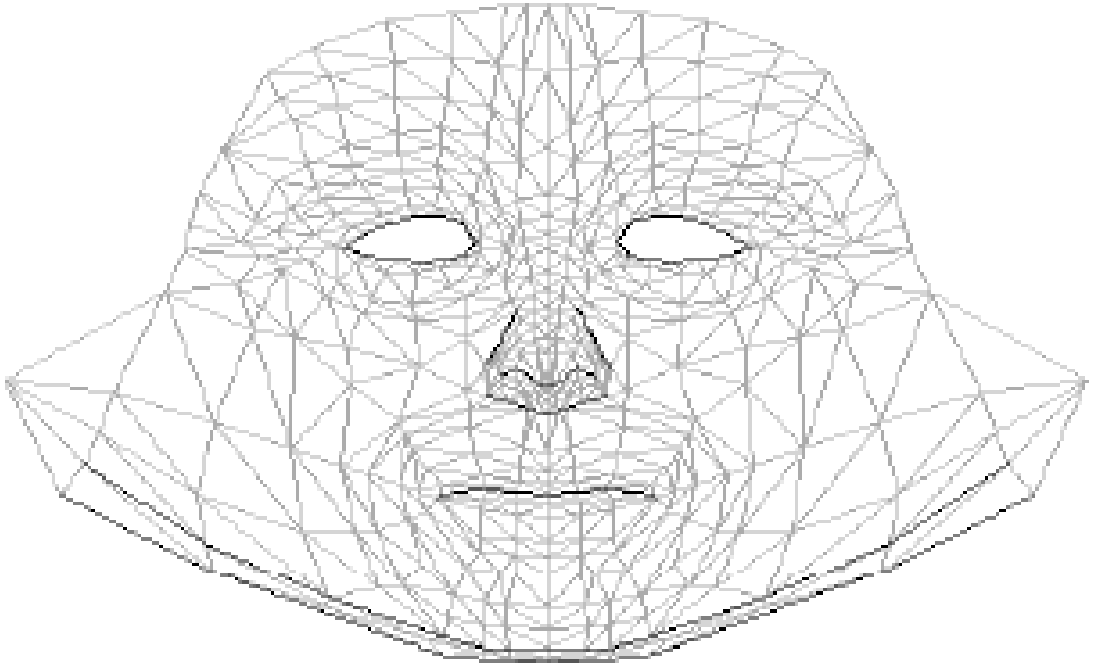
- A laser range scanner
  - Rotates 360 degrees around the subject
  - Laser stripes are projected on to the head
- The range data is obtained
- The color texture data is obtained at the same time





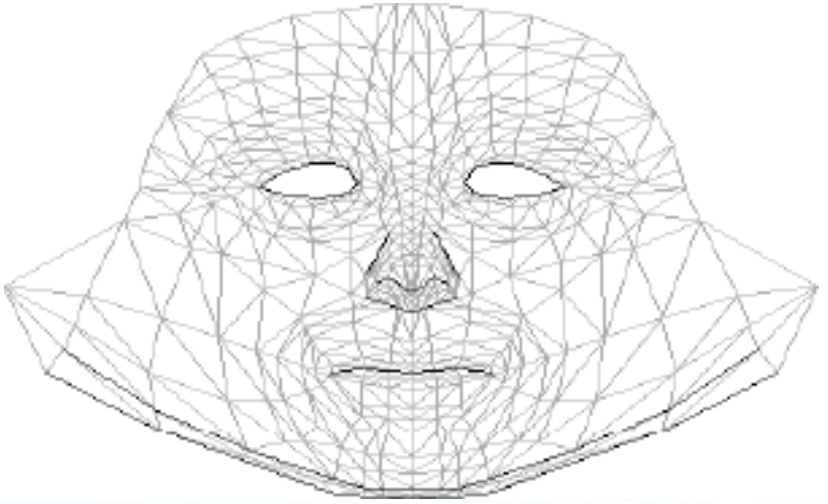

A Generic Face Mesh

- Reduce the large array of range data to a geometric surface model
- A generic face mesh is fit into the range data using feature points



Adaptation Procedure

- 1. Locate nose tip
- 2. Locate chin tip
- 3. Locate mouth contour
- 4. Locate chin contour
- 5. Locate ears
- 6. Locate eyes
- 7. Activate spring forces







(a) (b)

Figure 2: Range image (a) and positive Laplacian of range magnitude (b).

Locate the nose tip:

– the highest range data point in the central area, and globally translate the face mesh to achieve correspondence with the tip of the nose.

Locate the chin

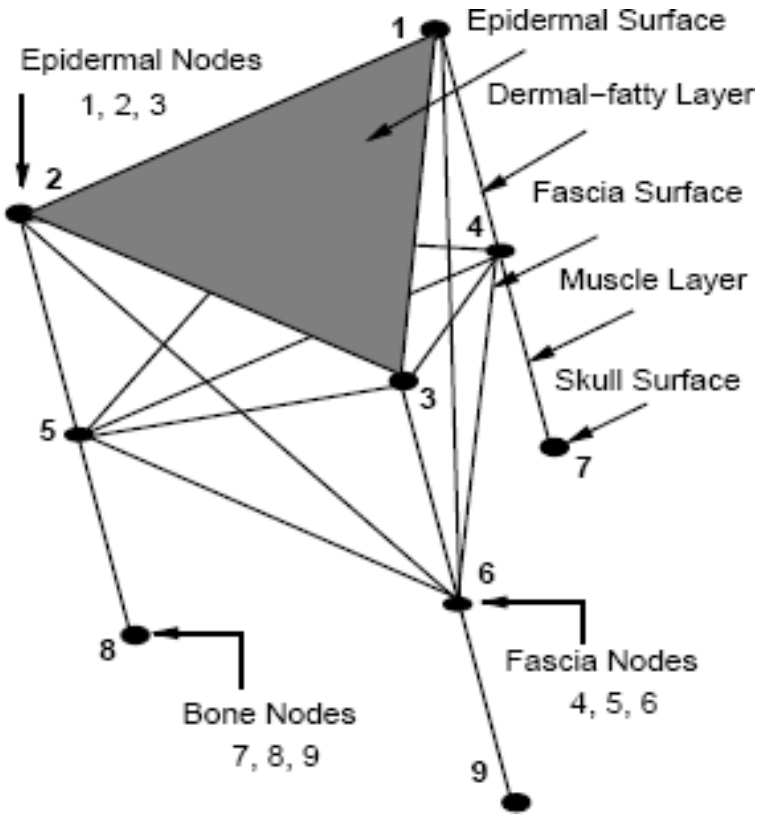
– the point below the nose with the greatest value of the positive Laplacian of range.

Locate the mouth

– the point of greatest positive Laplacian between the nose and chin

The Anatomical Model

- The face can be modeled by two layers and three surfaces
  - Dermal-fatty Layer
  - Muscle Layer
  - Epidermal surface
  - Fascia Surface
  - Skull Surface



Point-Mass System

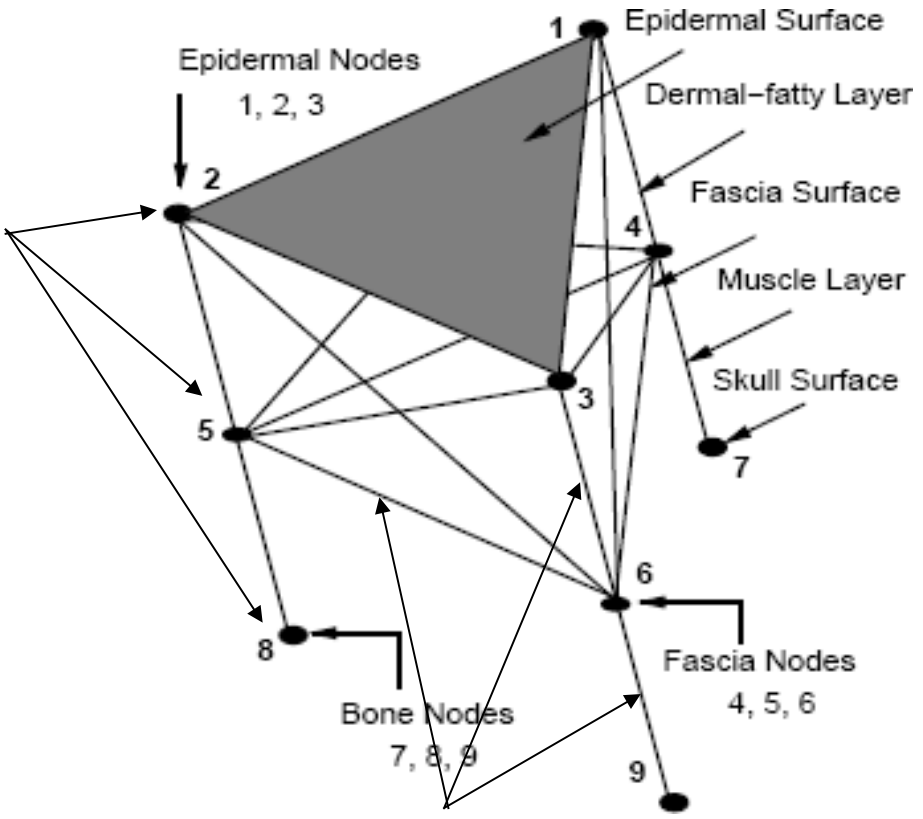
- The skin, fat and muscles are emulated by point masses connected by springs

elasticity direction

$$\mathbf{g}_j = c_j(l_j - l_j^r)\mathbf{s}_j$$

force Length Slack length

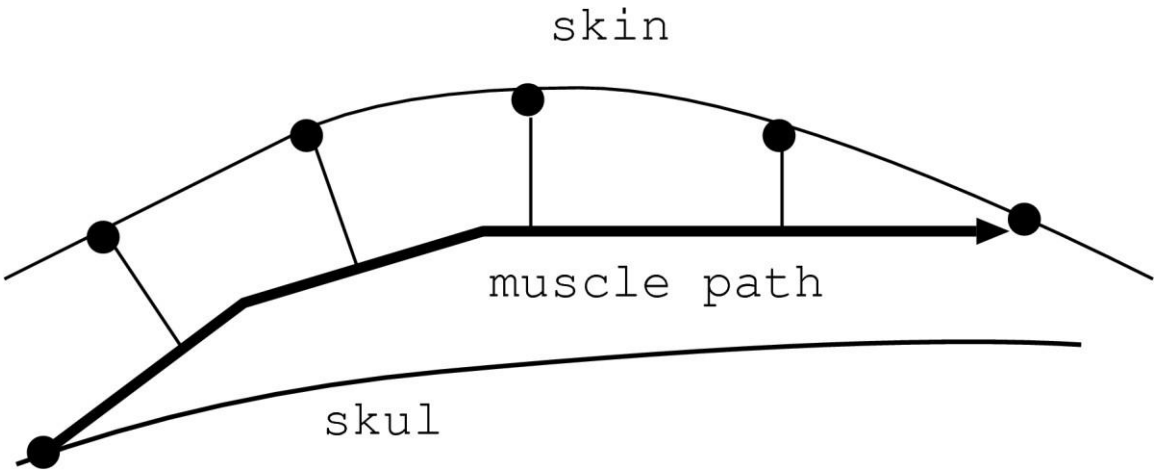
Point mass



springs

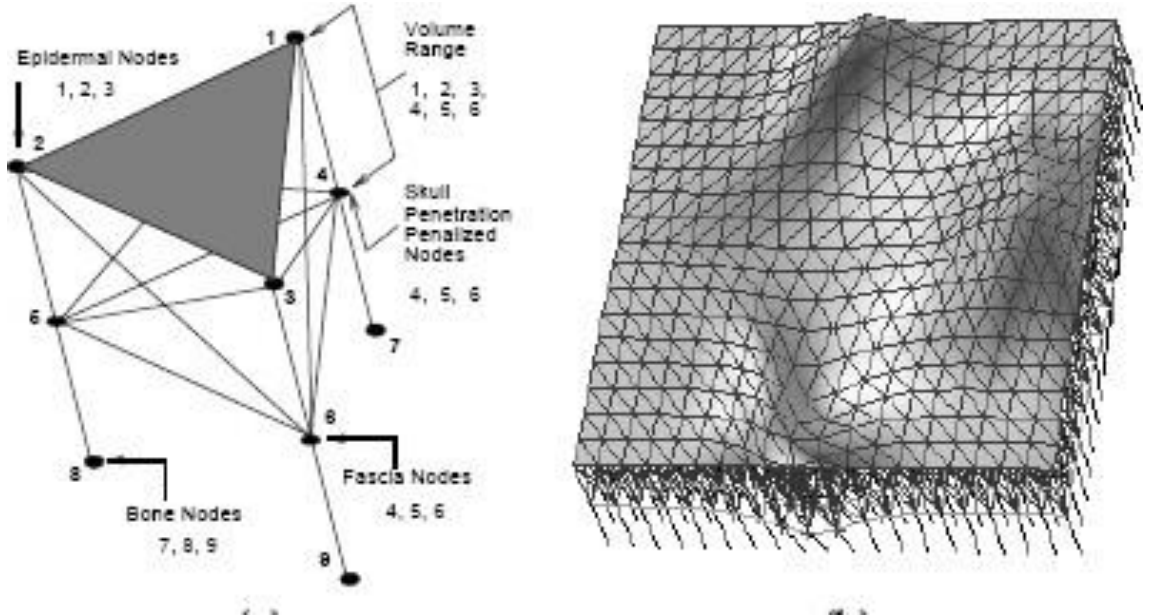
Muscle-Skin Attachment

- The muscle is connected to the skin at multiple points along its path
- Deforming the skin at multiple sites / not only at its end



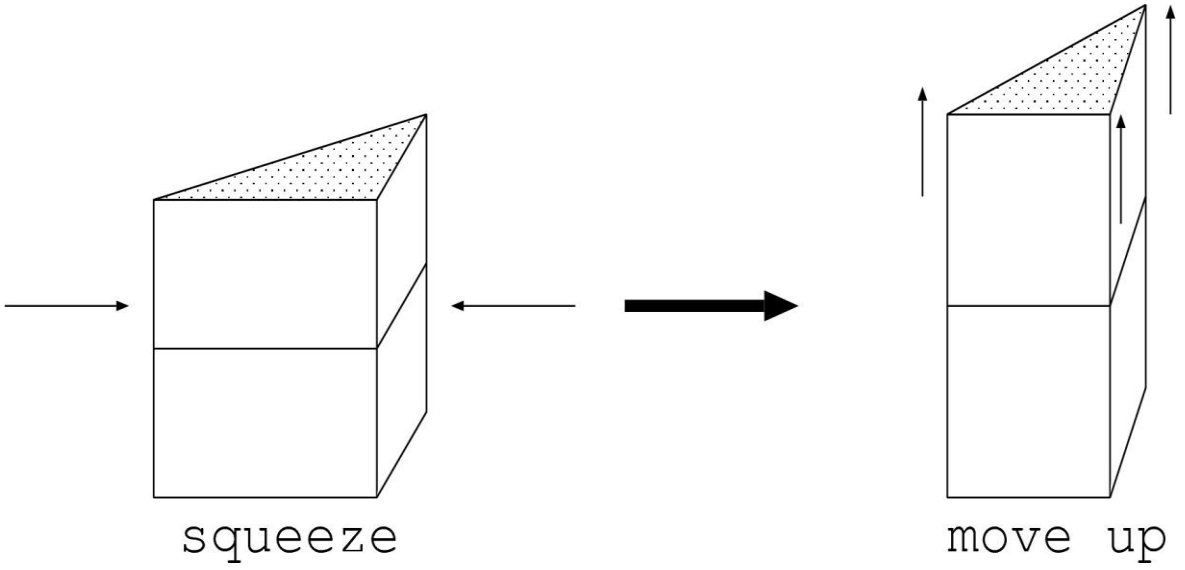
The Volume Preservation Forces

- The human skin is incompressible
- Volume preservation force is needed to simulate the wrinkles
- Pressing the node upwards proportionally to the decrement of the volume



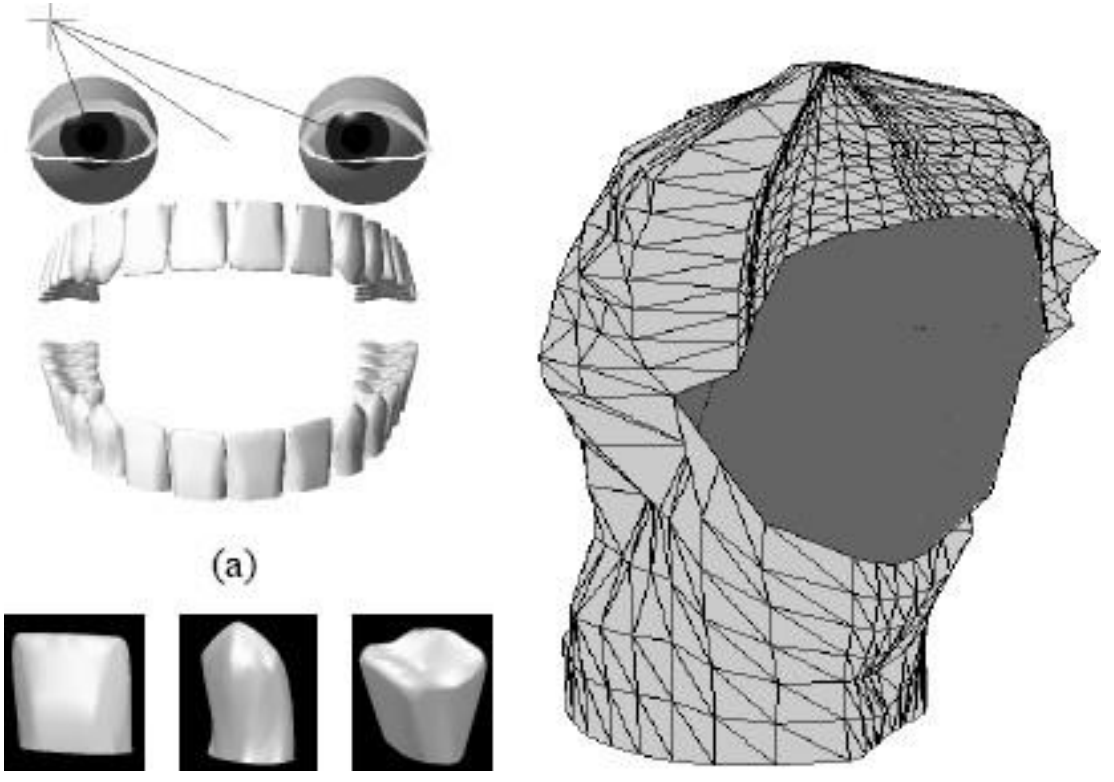
The Volume Preservation Forces

- The human skin is incompressible
- Volume preservation force is needed to simulate the wrinkles
- Pressing the node upwards proportionally to the decrement of the surface area



Geometry models for other head components

- Teeth, eyes, and neck need to be modeled separately
  - These data are difficult to be captured by the scanner



From Lee and Terzopoulos



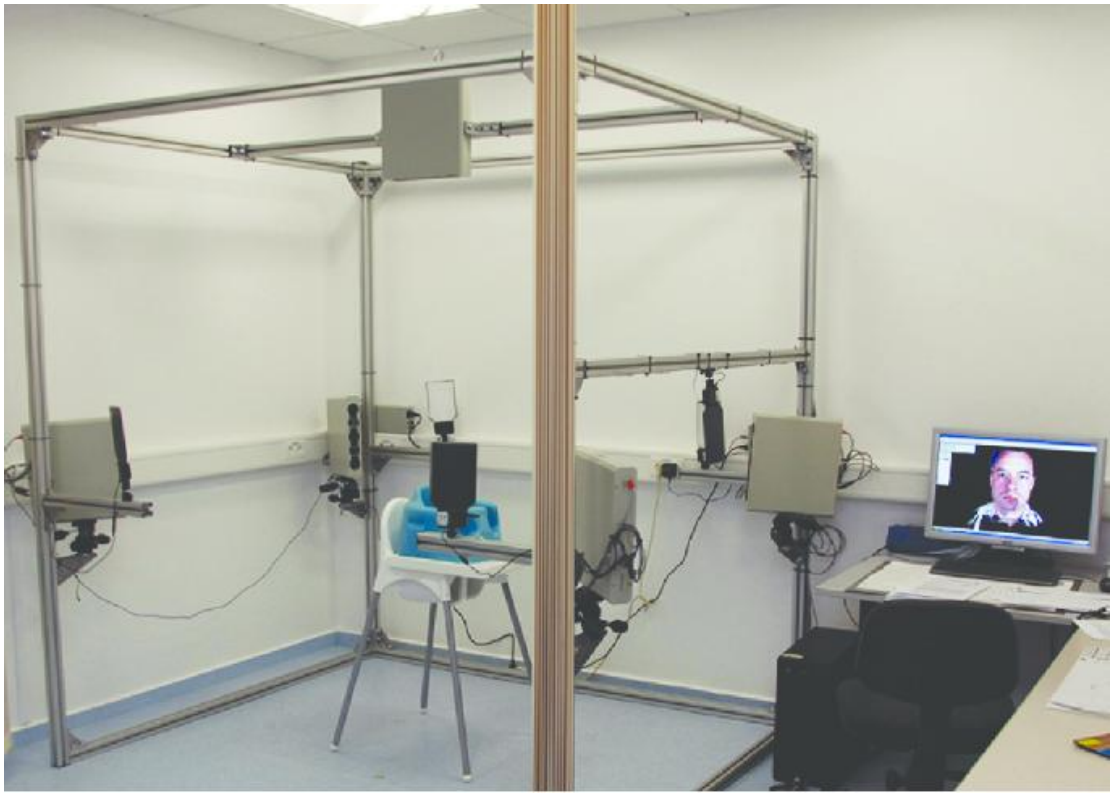
http://www.youtube.com/watch?v=dQef4pM_vXU

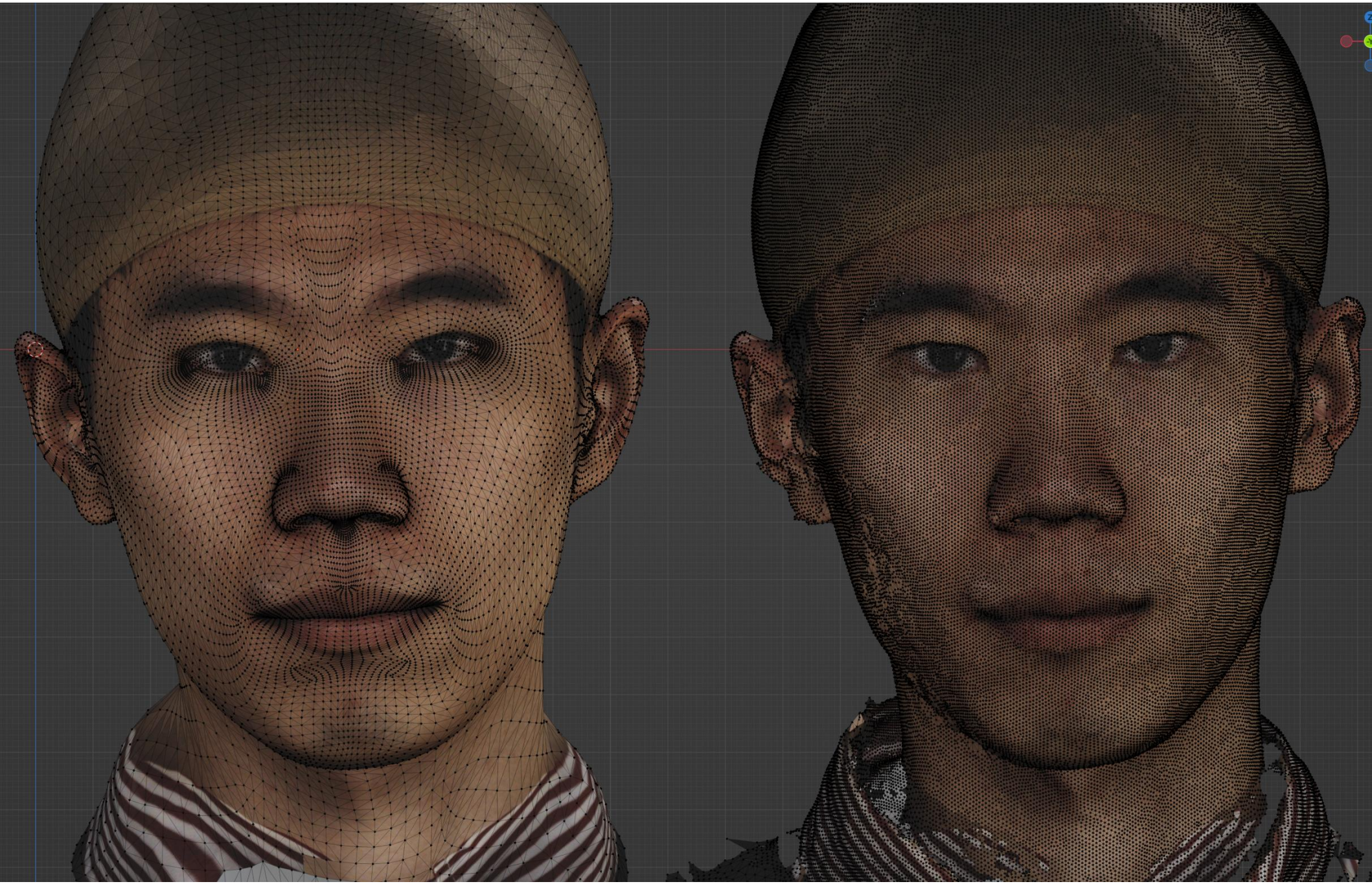


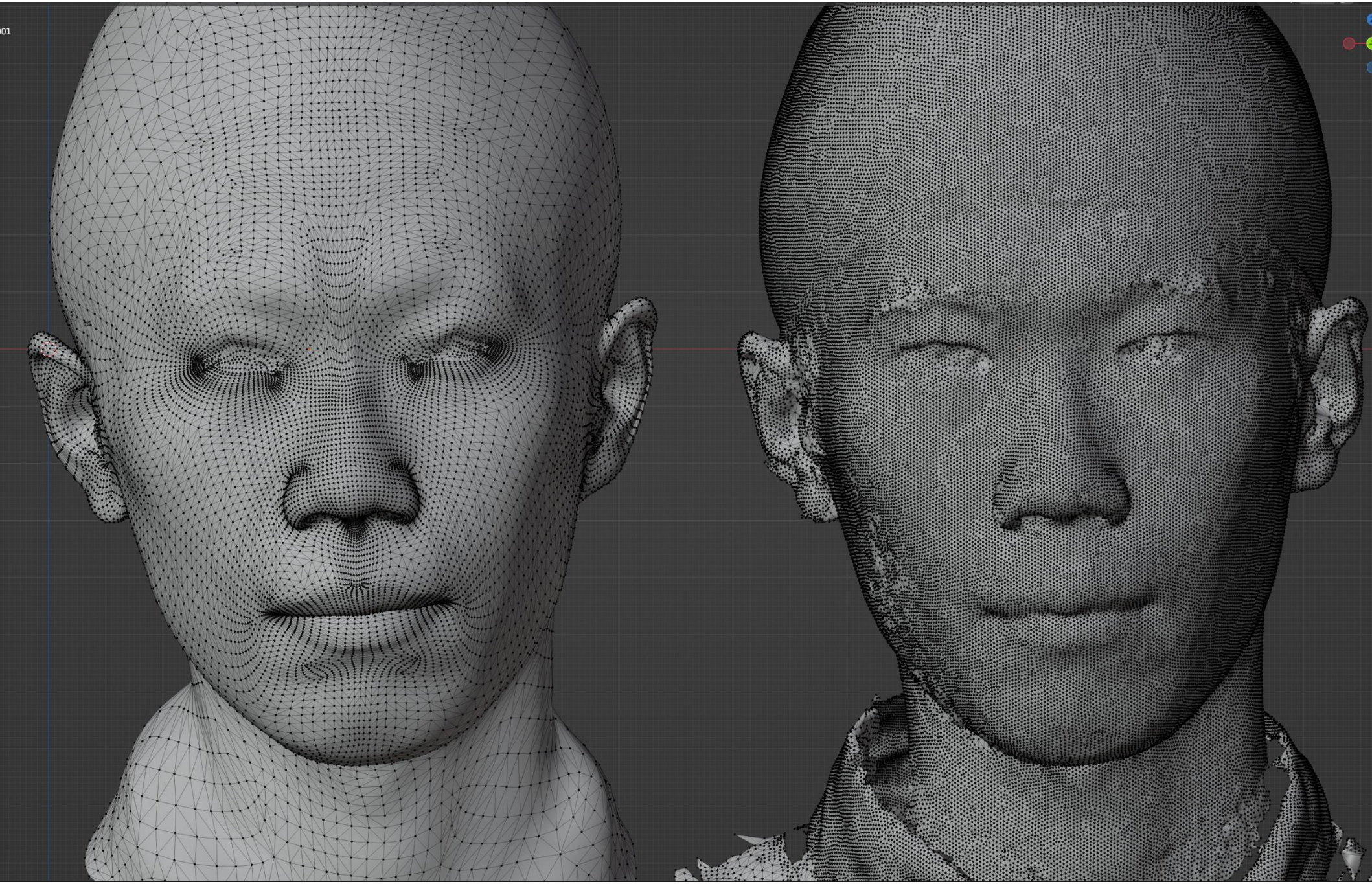


3dMD Face Scan System

A face scanner composed of several cameras
Can capture the geometry as well as texture







Muscle-based animation

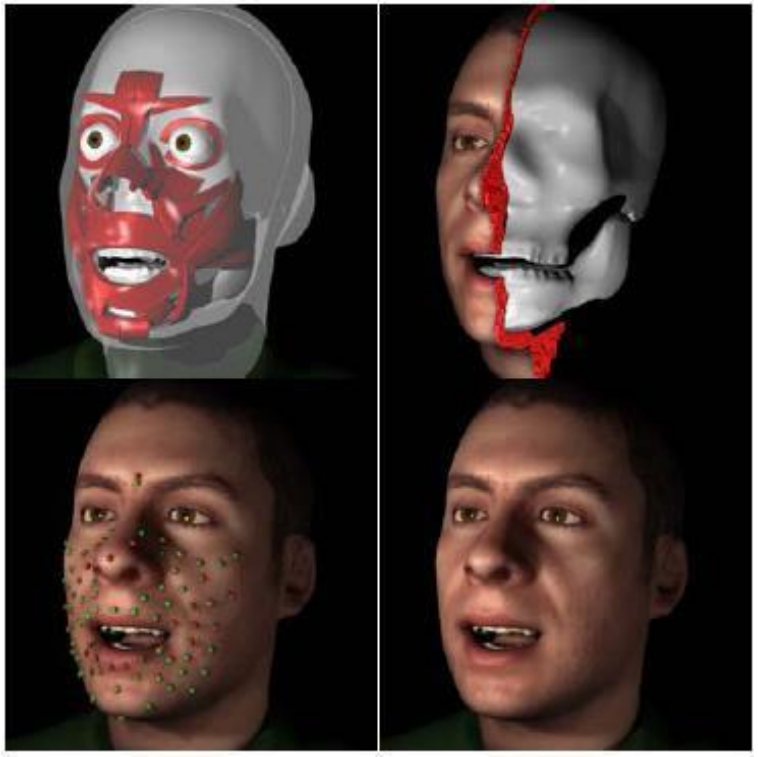
- Estimating the muscle activation from the motion capture data

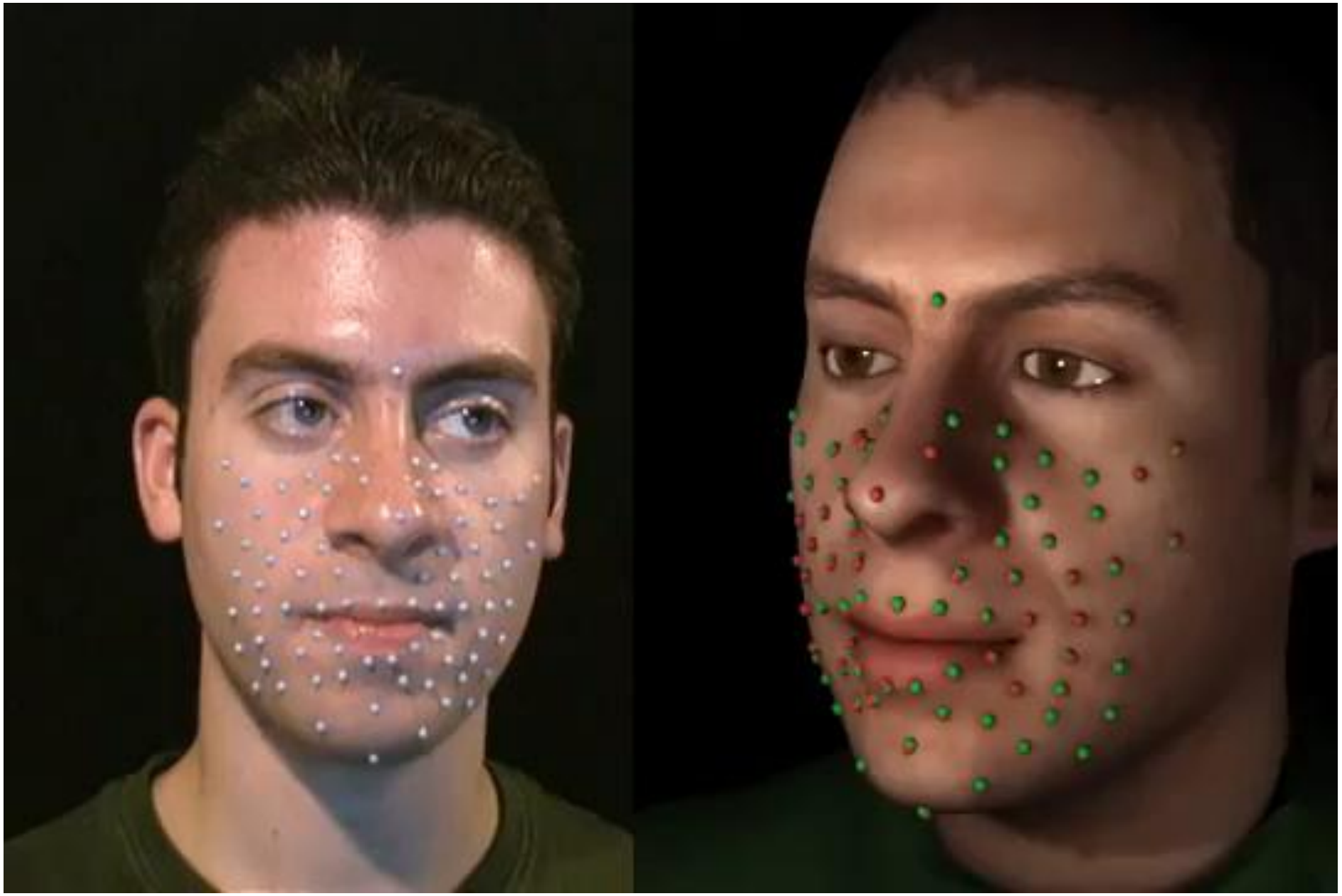
"Automatic determination of facial muscle activations from sparse motion capture marker data", SIGGRAPH 2005

•First, a precise anatomical model is Produced from Visible Human Motion Dataset

•Next, the muscles are activated so that the simulated location of the marker overlaps with its real location

http://www.youtube.com/watch?v=9dkhxgt5QeQ



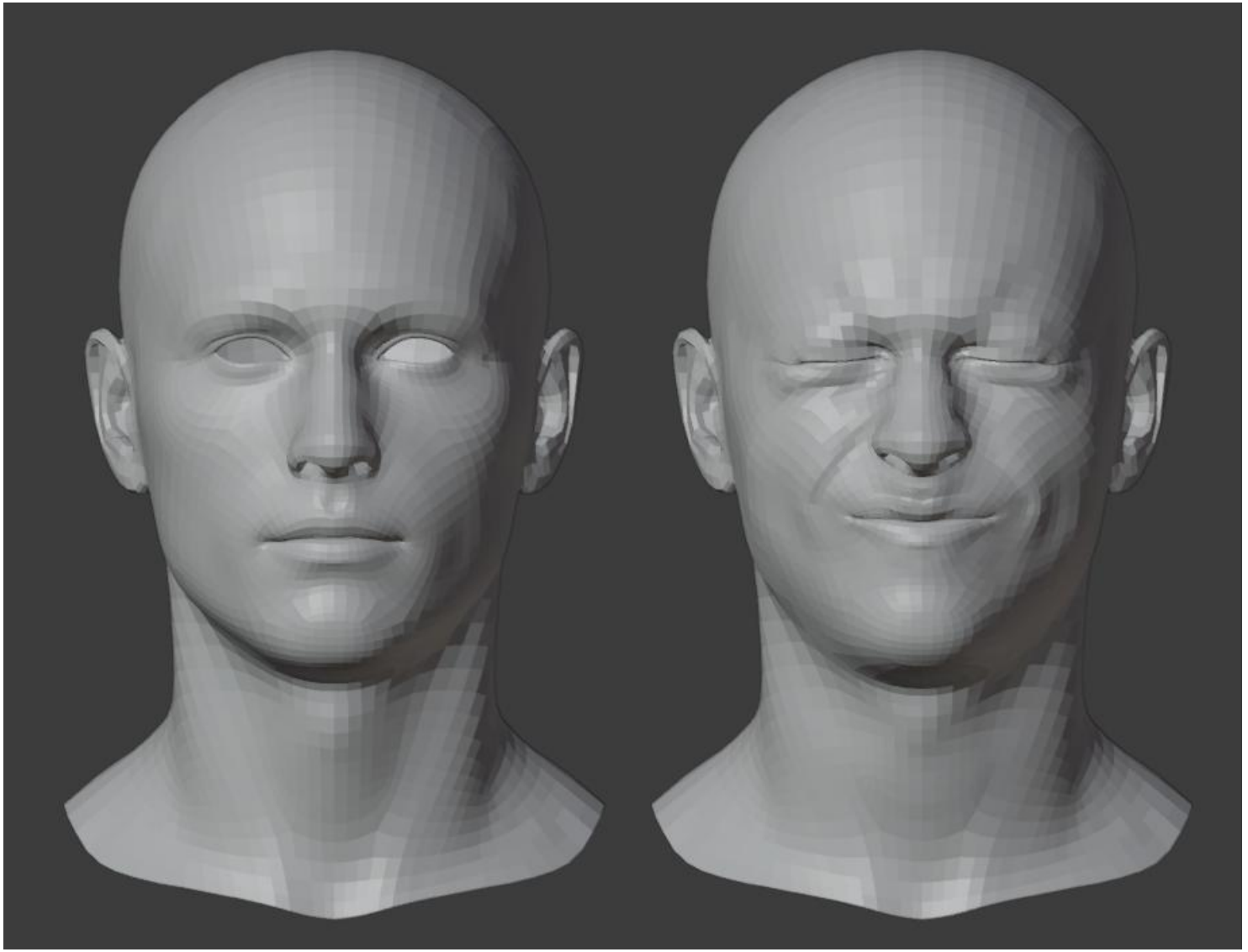


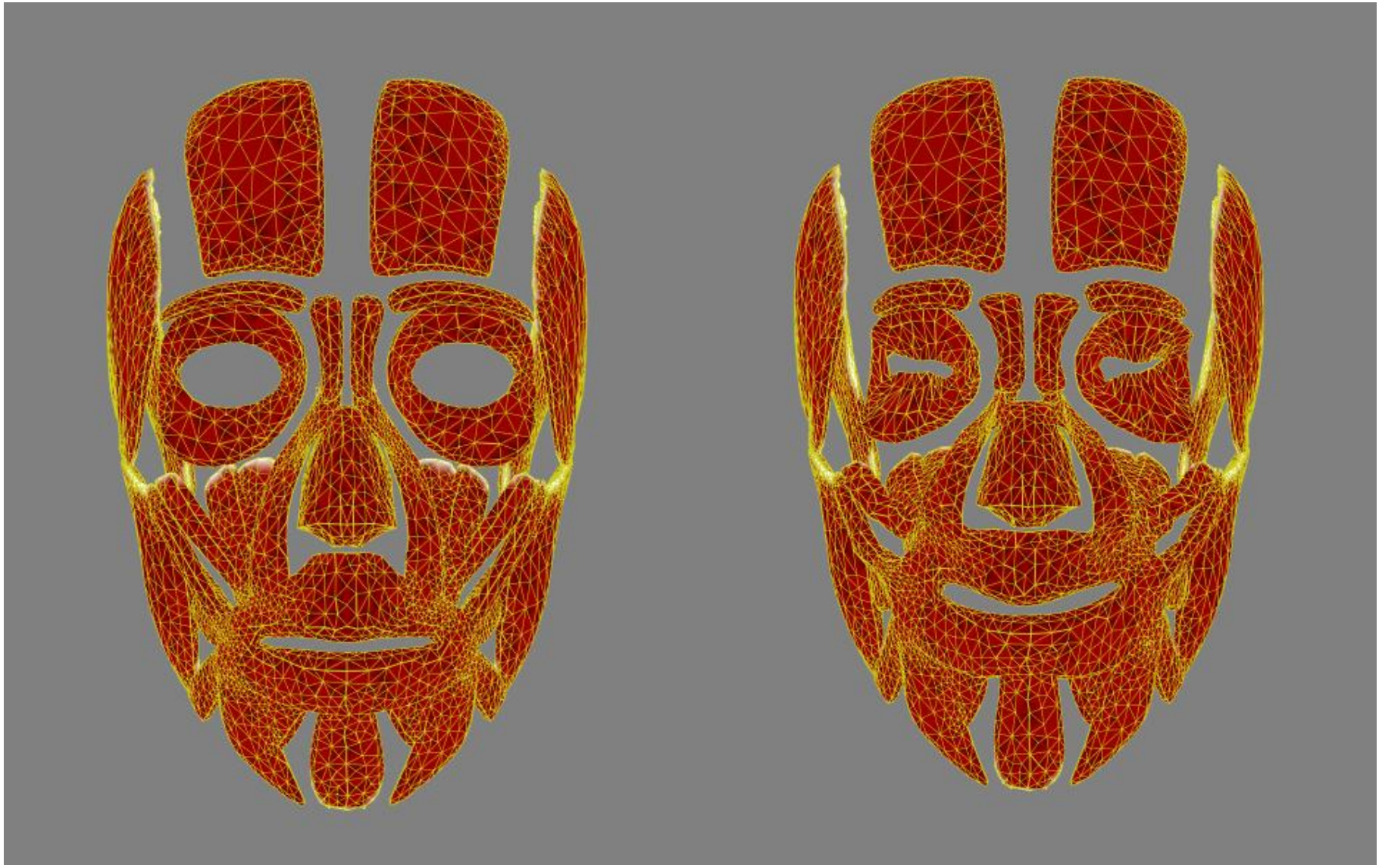
Difficulty with Musculoskeletal Models

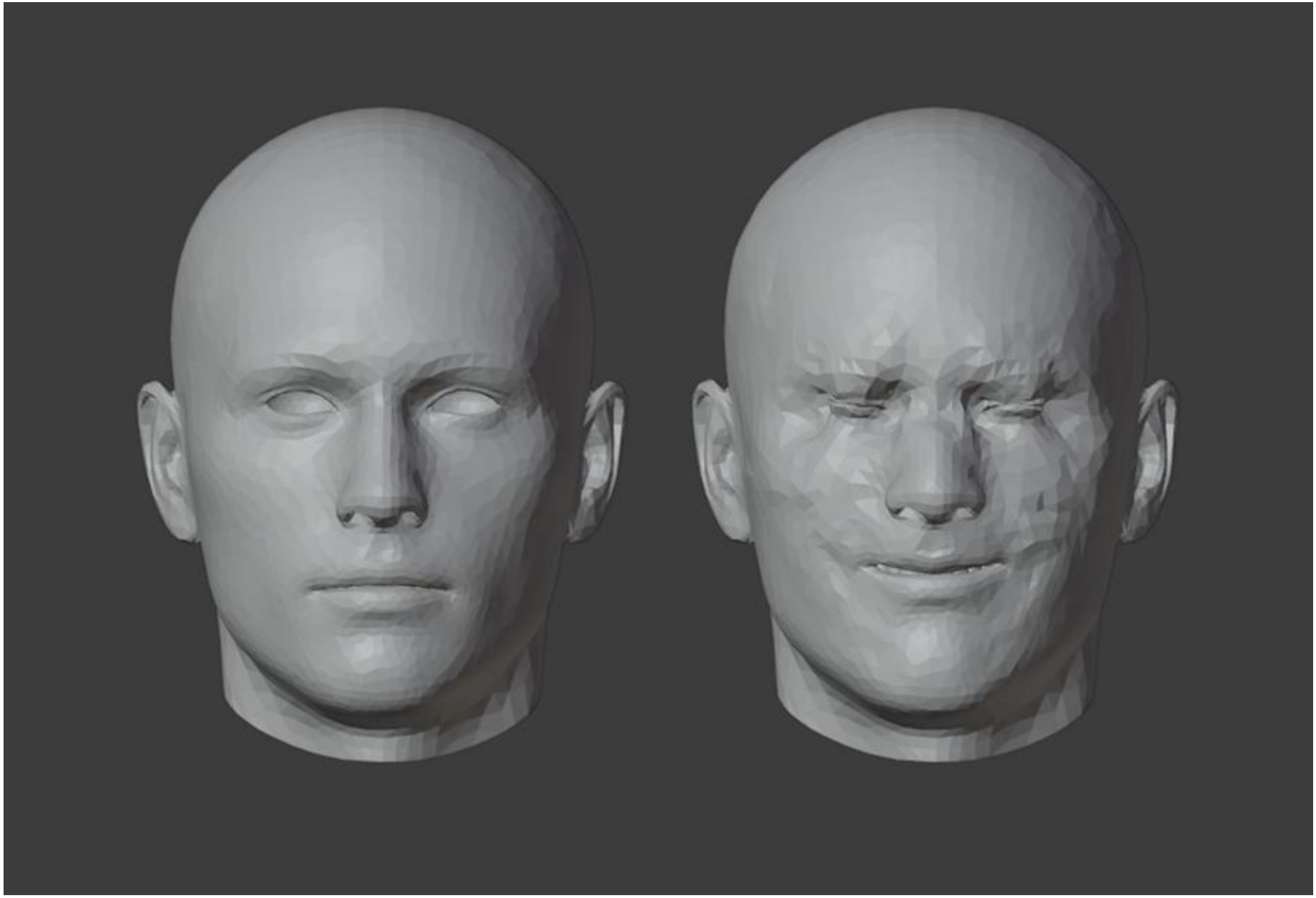
The quality of the results depend a lot on the quality of the anatomical model

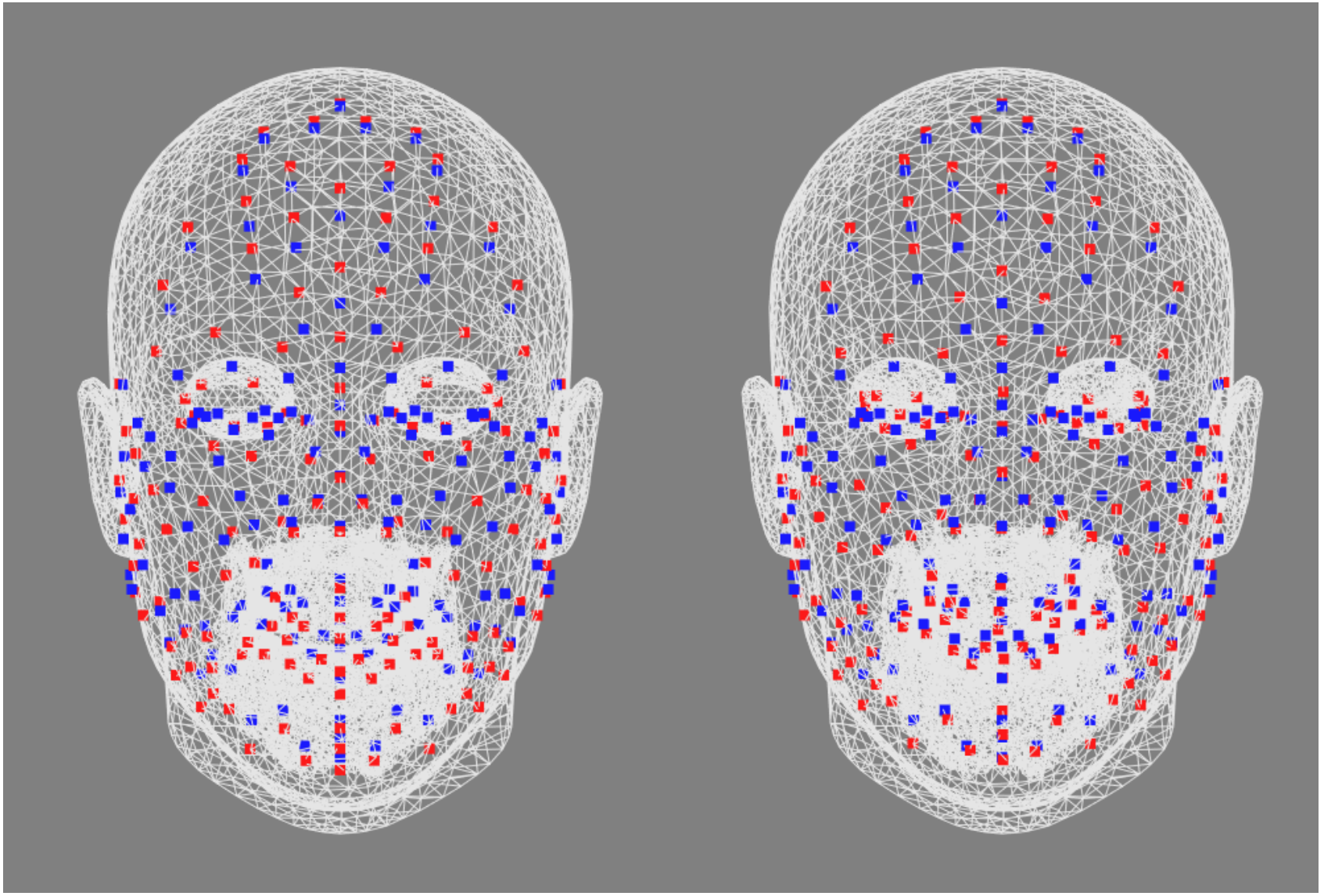
- The details of the muscles
  - the shape and volume of the muscle,
  - fiber directions,
  - attachments to the bones
- Soft materials
  - Fat layer, skin layer
- Activation of the muscles

Failure in modeling such items will result in poor tracking



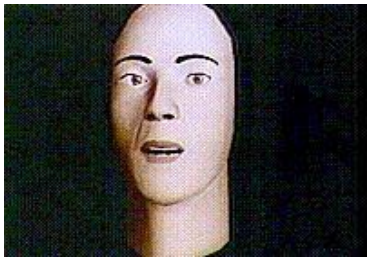


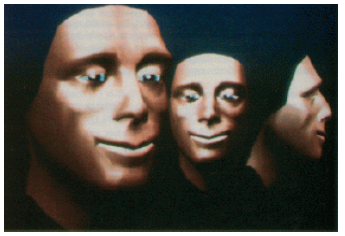




Overview

- Parke's parametric face
- Muscle-based models
- **Deformation Transfer**











Face motion capture

- We can capture the facial movements by using optical trackers, or simply tracking the features of the face





Face motion capture

- We can capture the facial movements by using optical trackers, or simply tracking the features of the face
- But the geometry of the virtual actor is different from the actor – how to control the virtual actor





Face motion capture

- We can capture the facial movements by using optical trackers, or simply tracking the features of the face
- But the geometry of the virtual actor is different from the actor – how to control the virtual actor

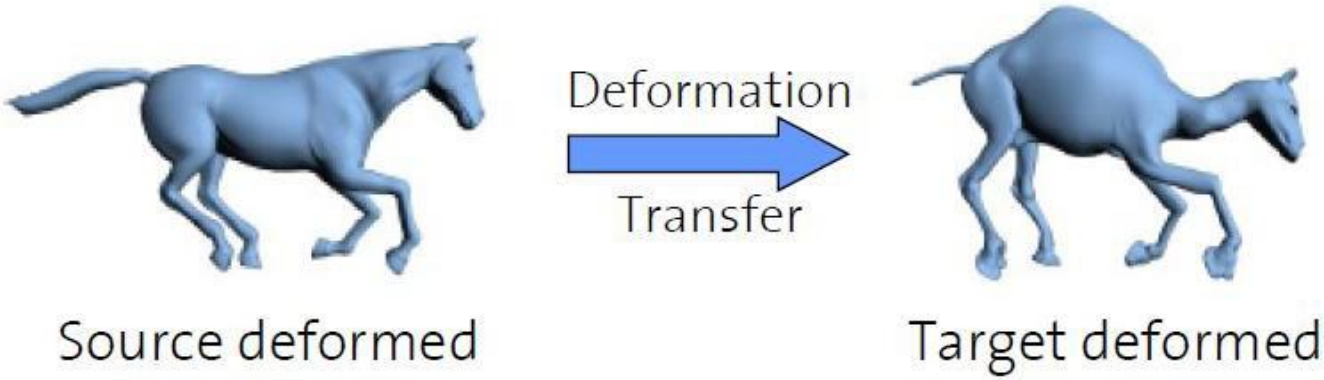
→ Deformation Transfer





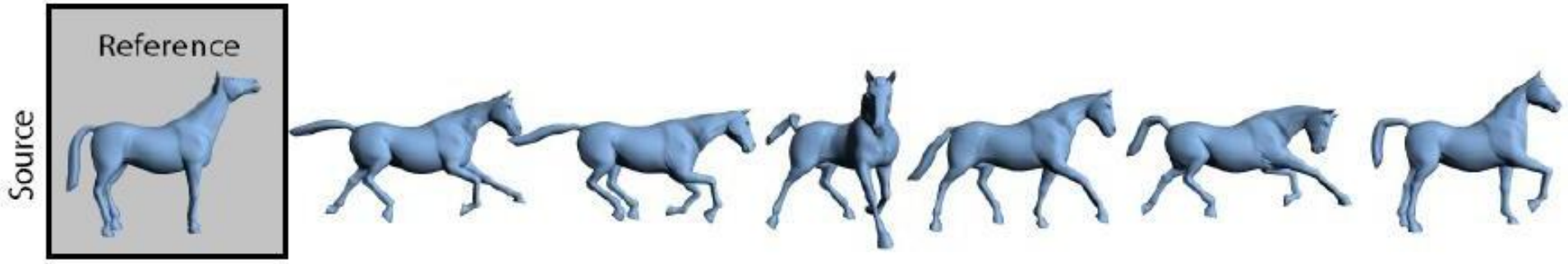
Deformation Transfer

- Deforming the source object to the target object
- Not targeting only faces, but arbitrary objects

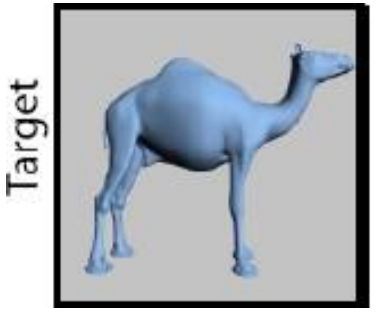


What is it about?

Given a source mesh in a reference pose and several deformations of it:

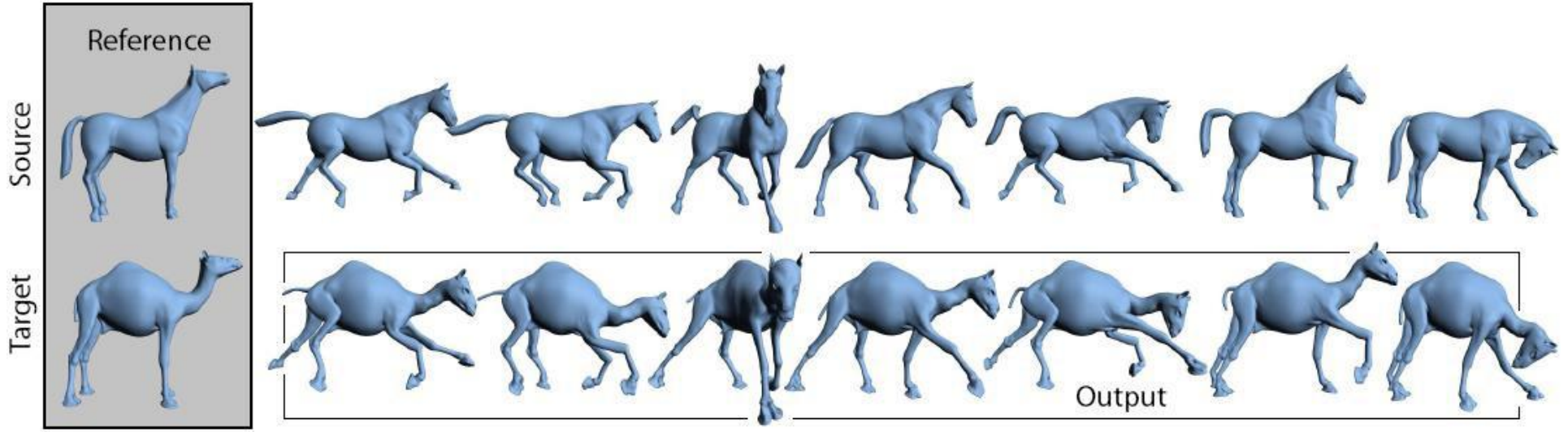


Given another mesh, called Target in same reference pose:



Deformation Transfer

- Transfer the deformation to the target

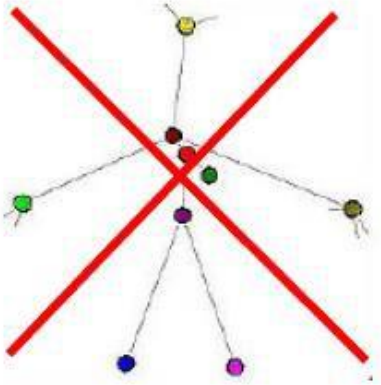


- Reuse of deformations which were probably created with a lot of effort

Deformation Transfer

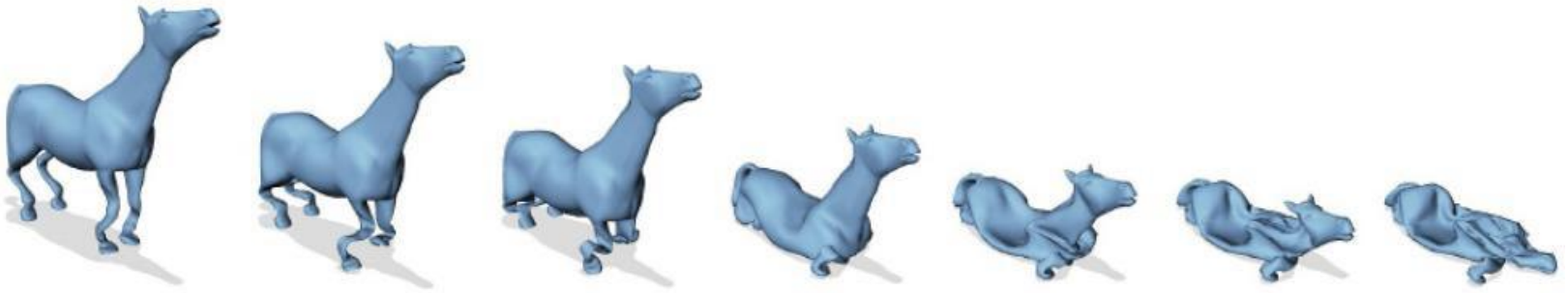
- Deforming Transfer is purely mesh-based
- No need of a skeleton





Deformation Transfer

- Not skeleton driven deformations
  - Applicable to non-rigid or facial deformations





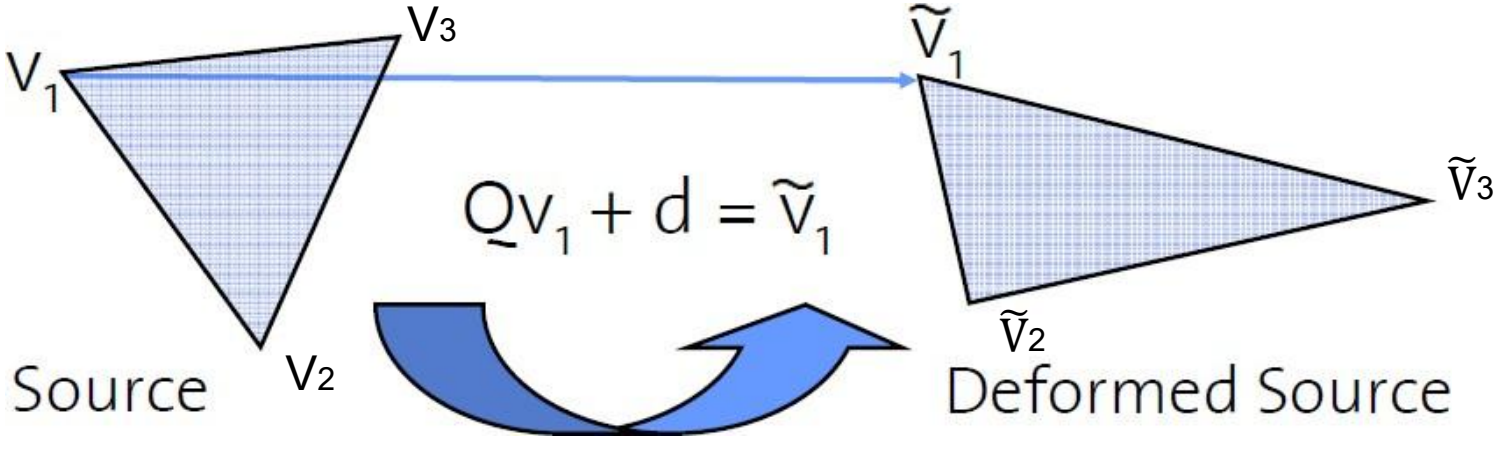
Approach

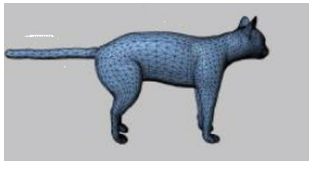
- Compute the deformation for every source triangle (orientation, scale, skew)
- Compute mapping from source to the target triangles (correspondence)
- Apply the deformation to the corresponding target triangles

Deformation Details

- Deformation based on per-triangle transformation

$$\mathbf{Q}\mathbf{v}_\mathbf{i} + \mathbf{d} = \tilde{\mathbf{v}}_\mathbf{i} \quad i = 1 \ldots 4$$



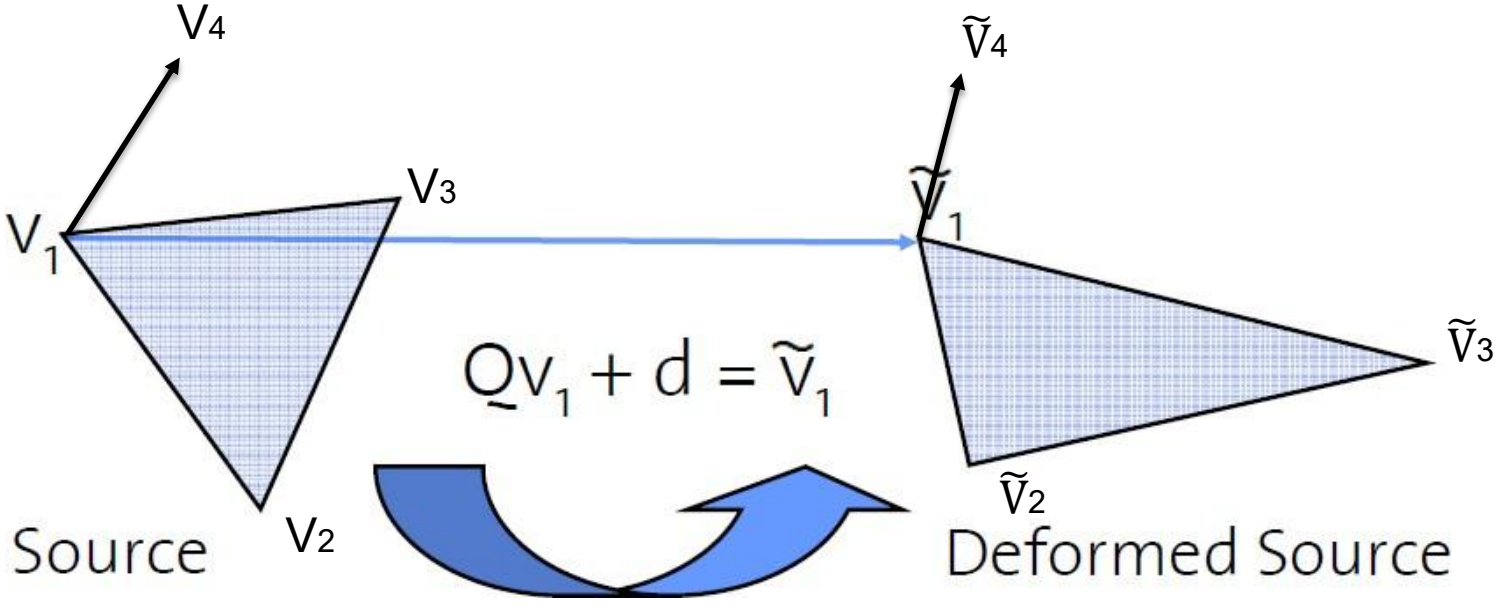


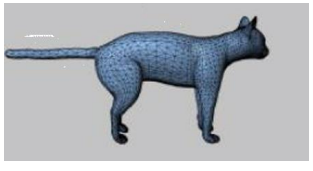


Deformation Details

- Deformation based on per-triangle transformation

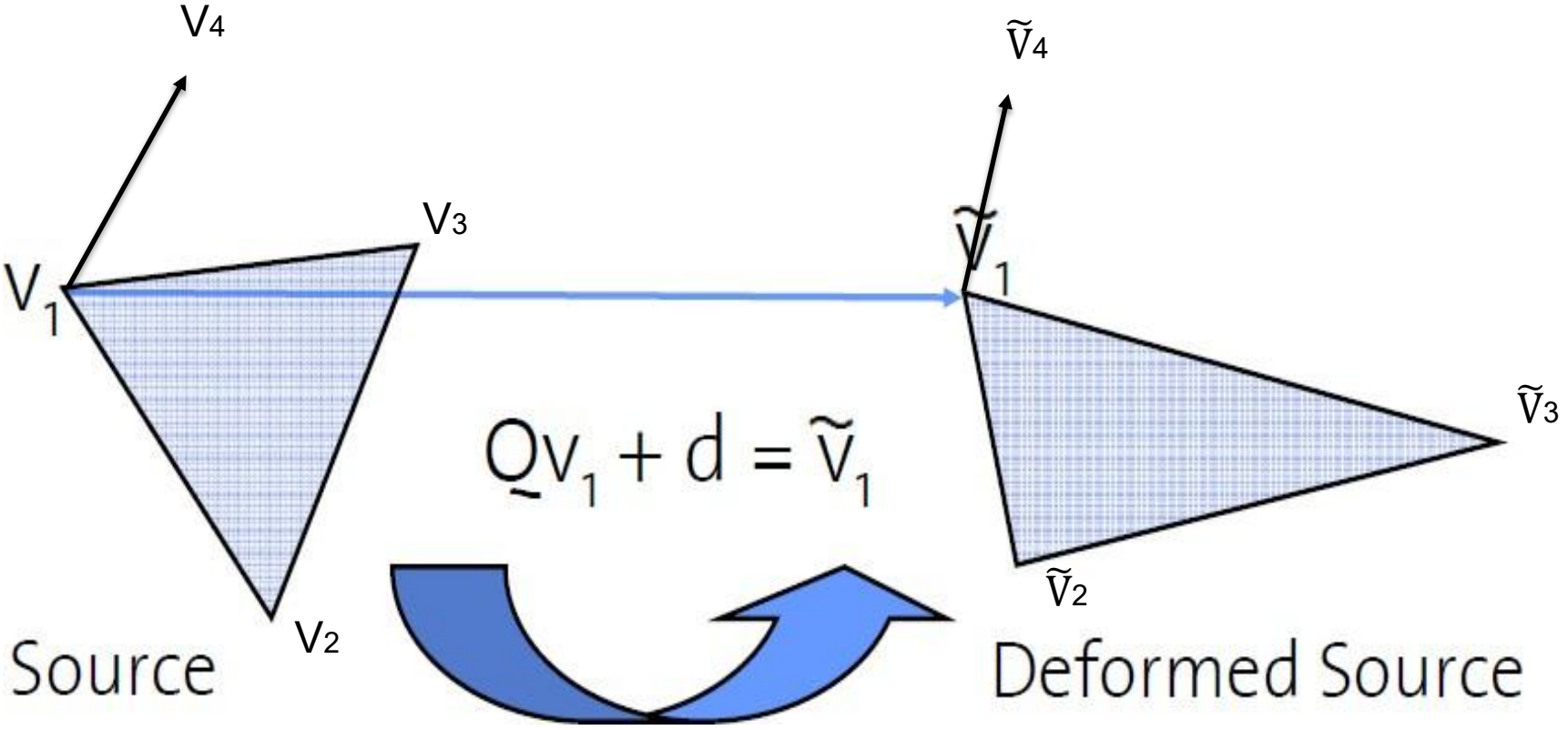
$$\mathbf{Q}\mathbf{v}_\mathbf{i} + \mathbf{d} = \tilde{\mathbf{v}}_\mathbf{i} \quad i = 1 \ldots 4$$







$$\mathbf{v}_{i_4} = \mathbf{v}_{i_1} + \frac{(\mathbf{v}_{i_2} - \mathbf{v}_{i_1}) \times (\mathbf{v}_{i_3} - \mathbf{v}_{i_1})}{\sqrt{\left\|(\mathbf{v}_{i_2} - \mathbf{v}_{i_1}) \times (\mathbf{v}_{i_3} - \mathbf{v}_{i_1})\right\|}},$$

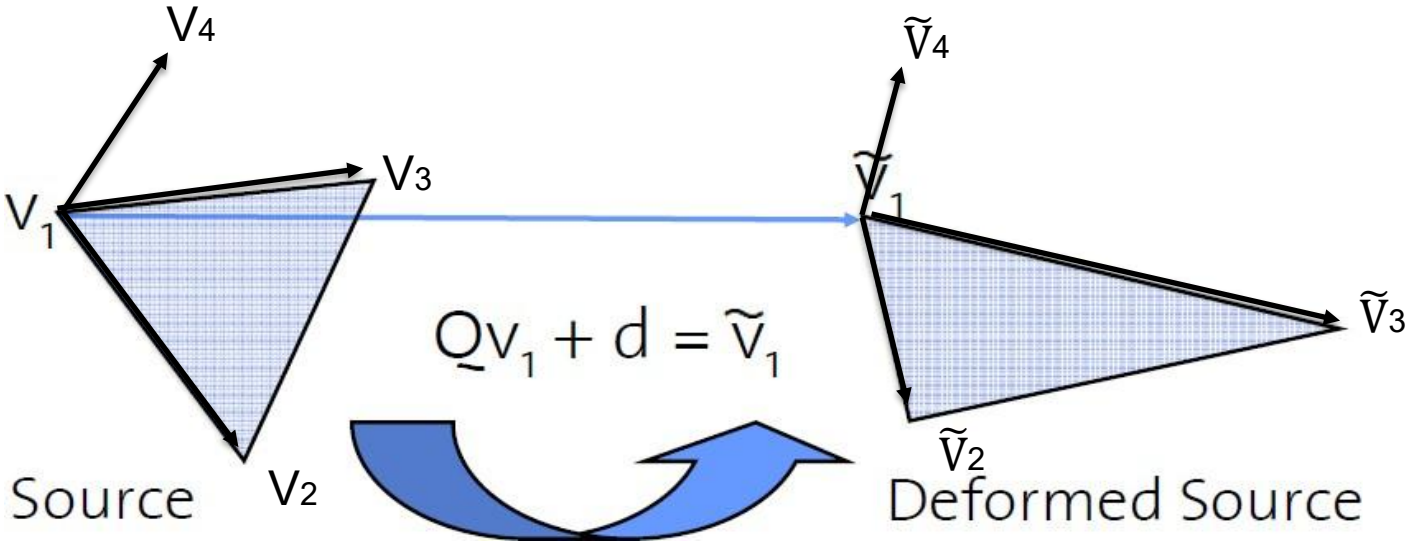


Computing the Q (Deformation Gradient)

$$\mathbf{V} = [\mathbf{v}_{i_2} - \mathbf{v}_{i_1} \quad \mathbf{v}_{i_3} - \mathbf{v}_{i_1} \quad \mathbf{v}_{i_4} - \mathbf{v}_{i_1}]$$

$$\tilde{\mathbf{V}} = [\tilde{\mathbf{v}}_{i_2} - \tilde{\mathbf{v}}_{i_1} \quad \tilde{\mathbf{v}}_{i_3} - \tilde{\mathbf{v}}_{i_1} \quad \tilde{\mathbf{v}}_{i_4} - \tilde{\mathbf{v}}_{i_1}]$$

$$\mathbf{Q} = \tilde{\mathbf{V}}\mathbf{V}^{-1}$$

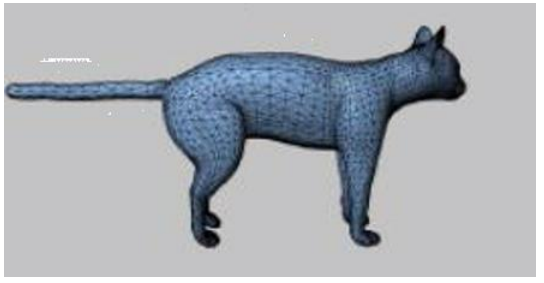


Closed Form Solution

- Closed form solution of Q:

$$\mathbf{Q} = \tilde{\mathbf{V}}\mathbf{V}^{-1}$$

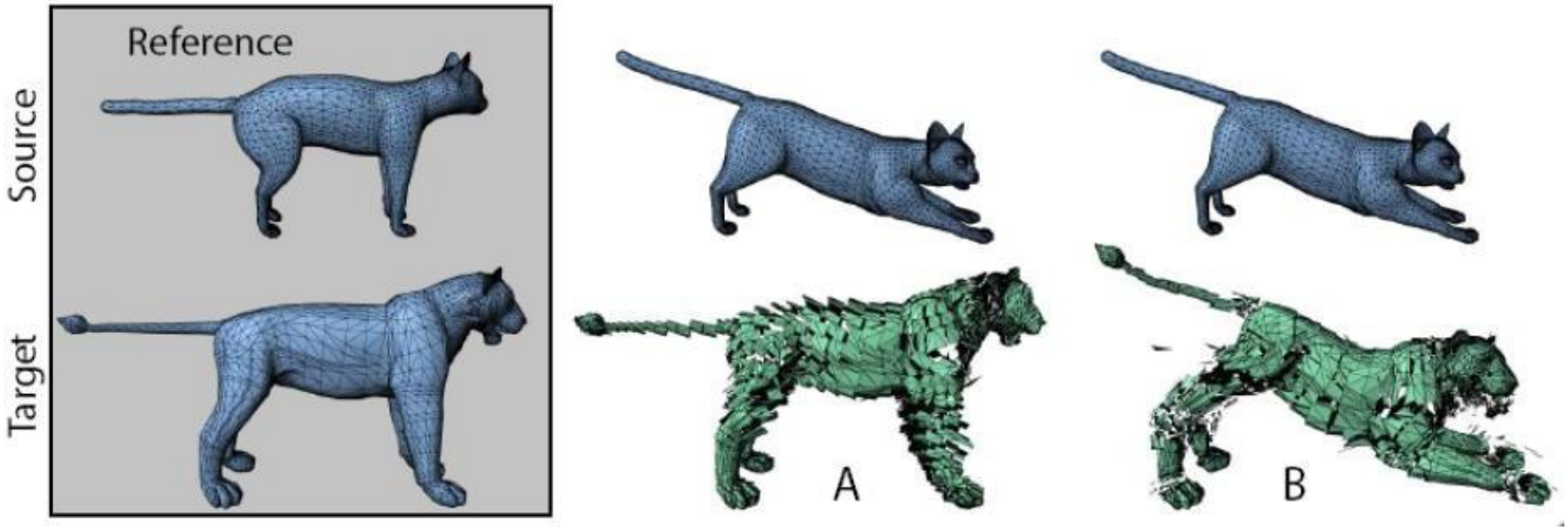
- "Deformation Gradient" Q depends on
  - Triangle in reference pose
  - Triangle in deformed pose





Applying Deformation Gradient to the Target Mesh

- Leads to holes in the resulting mesh
- Used representation affords too many degrees of freedom



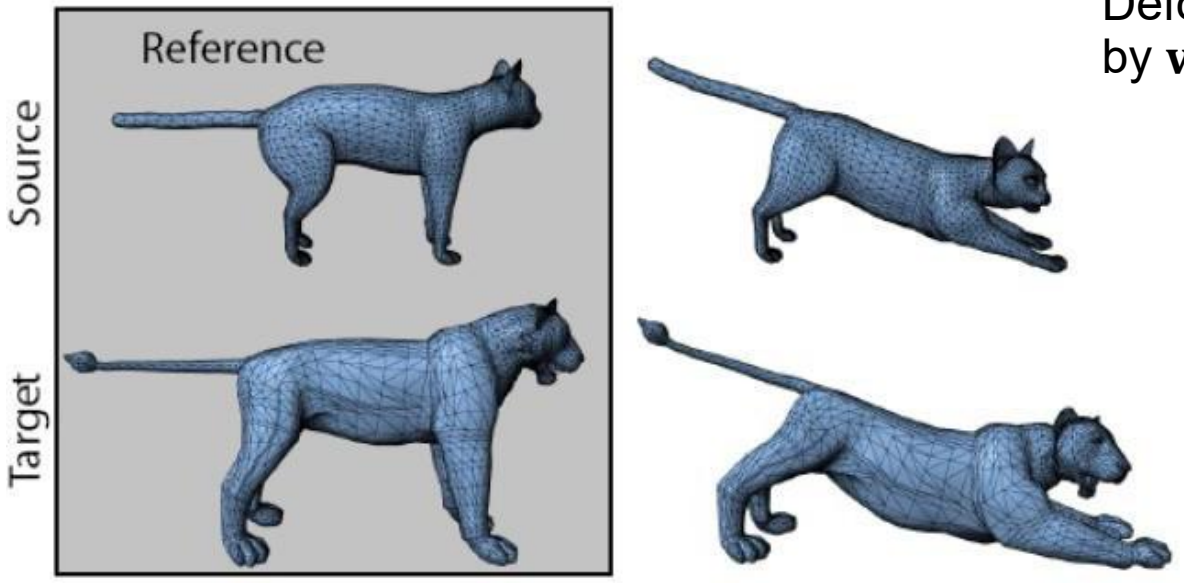
Minimization Problem

- Preservation of consistency leads to an optimization problem

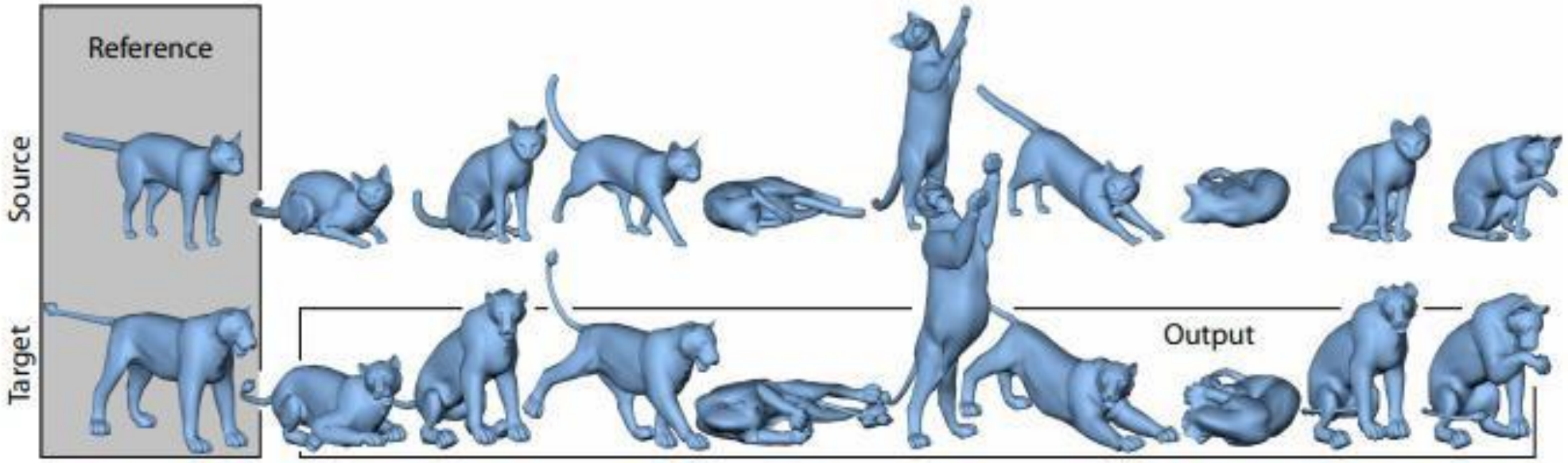
$$\min_{\tilde{\mathbf{v}}_1 \dots \tilde{\mathbf{v}}_n} \sum_{j=1}^{|M|} \left\| \mathbf{S}_{s_j} - \mathbf{T}_{t_j} \right\|_F^2$$

Deformation gradient by $\mathbf{v_1}$, … , $\mathbf{v_n}$

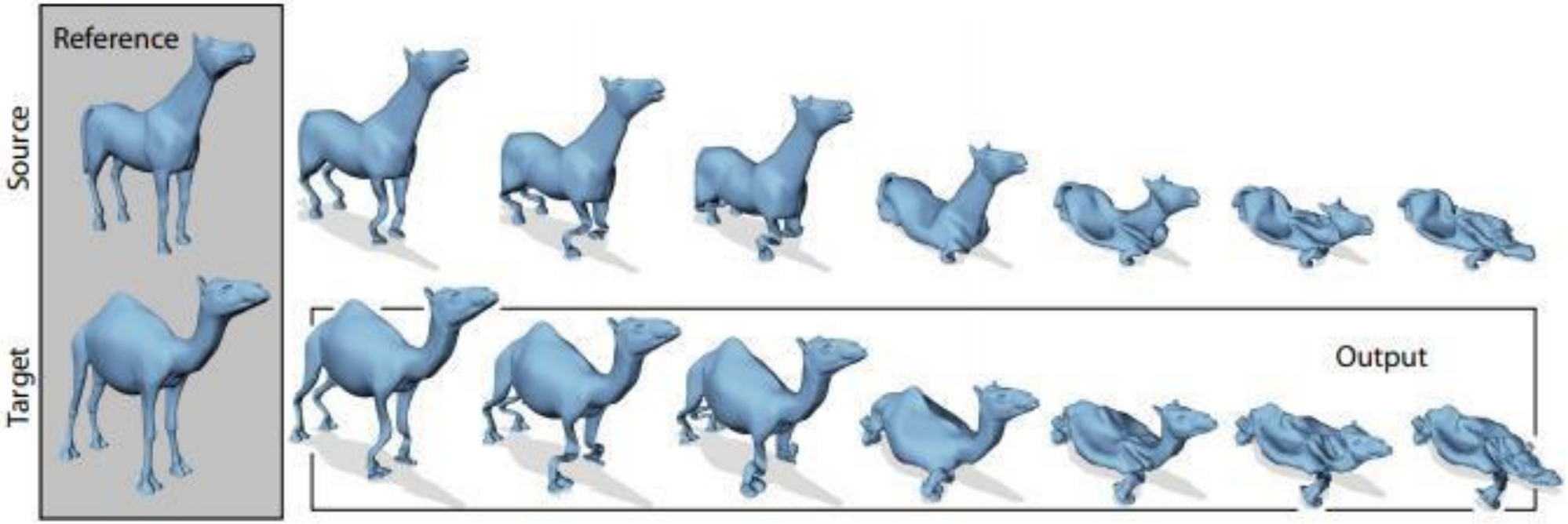
Deformation gradient of the target



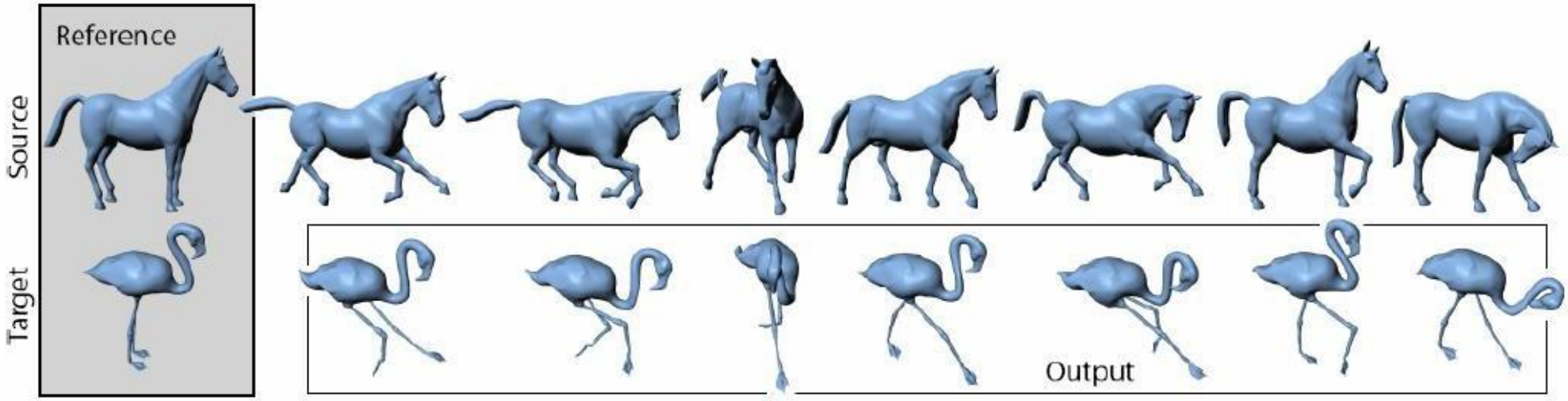
Results : Cat to Tiger



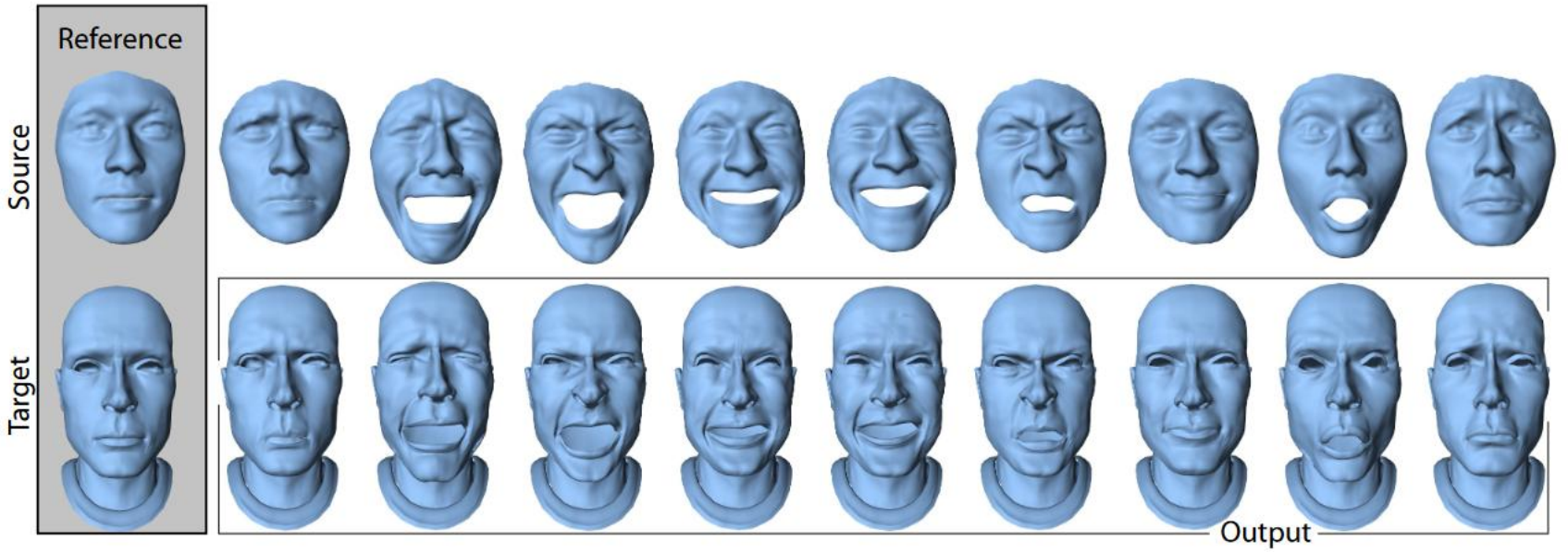
Results : Horse to Camel



Results : Horse to Flamingo



Human Face Transfer



Deformation Transfer for Triangle Meshes

Robert W. Sumner
Jovan Popović

Deformation Transfer of faces



Summary

- Deformation Transfer
  - Standard approach for transferring geometry to another shape
  - Applicable to faces as well as full-body characters

Readings

Yuencheng Lee, Demetri Terzopoulos, and Keith Waters, **Realistic Modeling for Facial Animation** SIGGRAPH '95

K. Waters. A muscle model for animating three-dimensional facial expression. *Computer Graphics*, 22(4):17–24, 1987.

- Sifakis et al. **Automatic determination of facial muscle activations from sparse motion capture marker data**, SIGGRAPH 2005
- Ekman P and Friesen W "Manual for the the Facial Action Coding System" *Consulting Psychologist* 1977 Press Palo Alto California
- **Expression Cloning,** Jun-yong Noh Ulrich Neumann SIGGRAPH 2001
- **Park et al., A feature-based approach to facial expression cloning CASA2005**
- **Deformation Transfer for Triangle Meshes, Sumner et al. SIGGRAPH 2004**
- Neural Face Rigging for Animating and Retargeting Facial Meshes in the Wild , Qin et al. SIGGRAPH 2023